Master's Thesis Proposal

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31 May 2026

Contents

1 Background and Motivation

1.1 Video Super-Resolution for Long Videos

Video super-resolution (VSR) has advanced rapidly with the adoption of diffusion-based and transformer-based architectures. Where earlier CNN-based methods upscaled fixed short clips (typically 7–32 frames), modern methods such as MGLD-VSR [Yang et al., 2023] and Upscale-A-Video [Zhou et al., 2023] can process substantially longer sequences in latent or sliding-window fashion, enabling deployment on real-world footage that runs into minutes rather than seconds.

This shift toward long-video SR introduces an evaluation gap. The standard VSR benchmarks — REDS (100-frame clips at 720×1280), Vimeo-90K (7-frame septuplets), Vid4, UDM10, and SPMCS — were designed for short-clip evaluation and report full-reference metrics (PSNR, SSIM, LPIPS) against paired HR ground truth. When applied to long videos in the wild, three fundamental limitations emerge:

- **No HR ground truth.** Real deployment videos have no paired high-resolution reference, making full-reference metrics (PSNR, SSIM, LPIPS, DISTS) inapplicable.
- **Adjacent-frame temporal metrics miss long-range artefacts.** Existing temporal metrics — tOF and tLP from TecoGAN [Chu et al., 2020], and E*warp from DOVE [Chen et al., 2025] — are computed at a single adjacent-frame gap (k=1). A method that is locally smooth but drifts globally over minutes is undetectable by these measures.
- **Smoother-output bias in learned representations.** DINOv2-based consistency metrics and LPIPS favour methods whose outputs change less in feature space between frames. Diffusion-based methods produce detail-rich outputs that score worse on these metrics despite being visually superior, a structural bias that misrepresents quality for modern SR architectures.

This thesis addresses the evaluation gap: long-video SR (videos longer than one minute) requires no-reference metrics that capture temporal consistency across multiple time scales and are robust to the smoother-output bias of learned representations.

1.2 Existing Metric Landscape

The existing metric ecosystem can be organised into five families:

1. **Full-reference pixel/perceptual** (PSNR, SSIM, LPIPS, DISTS) — require HR ground truth; inapplicable to deployment-time long-video evaluation.
2. **No-reference single-aspect** (CLIP-IQA [Wang et al., 2022], MUSIQ [Ke et al., 2021], NIQE, BRISQUE) — measure per-frame quality; no temporal dimension.
3. **Temporal pixel-level** (tOF, tLP from TecoGAN; E*warp from DOVE) — adjacent-frame (k=1) optical-flow consistency; systematically miss long-range temporal structure.

4. **Semantic video quality** (VBench [Huang et al., 2023], VBench-2.0 [Huang et al., 2025]) — multi-aspect benchmarks covering subject consistency, Human_Identity, Human_Anatomy, and others. These benchmarks were designed for text-to-video generation evaluation; thirteen of VBench-2.0's eighteen dimensions require text prompts and are therefore inapplicable to super-resolution, while the remaining prompt-independent dimensions operate at clip level and average per-clip scores up to a single video number, removing the cross-clip drift signal that long-video consistency assessment requires. Content-dependent failure modes are documented in Section 5.

5. **Disentangled aesthetic-technical NR** (DOVER [Wu et al., 2022]) — a learned video quality predictor whose final score is a combination of two disentangled convolutional branches: an aesthetic branch trained on perception-driven labels and a technical branch trained on distortion-driven labels. DOVER is per-video and produces a single aggregate; it does not specifically model cross-frame temporal drift.

The thesis problem is that no existing metric satisfies all four desiderata simultaneously: (1) no-reference, (2) multi-time-scale temporal coverage, (3) robust to smoother-output bias from learned representations, and (4) reliable across diverse long-video content including close-up scenes.

2 Literature Review

2.1 Video Super-Resolution Methods

The two primary VSR methods evaluated throughout this thesis proposal represent the two dominant stylistic families in modern super-resolution:

- **MGLD-VSR** [Yang et al., 2023] (ECCV 2024) applies masked guided latent diffusion to progressively refine super-resolved frames. It is characterised by sharp, detail-preserving output and exact reproduction of published DOVE benchmark numbers on UDM10 (PSNR 24.23, SSIM and LPIPS to four decimal places), providing a reliable evaluation baseline.

- **Upscale-A-Video (UAV)** [Zhou et al., 2023] (CVPR 2024) uses a text-conditioned temporal diffusion model inspired by text-to-video generation. It produces smoother, perceptually cleaner output at the cost of some high-frequency detail. UAV reproduces DOVE UDM10 results within a consistent +1.33 dB PSNR offset attributable to a minor degradation pipeline configuration difference, confirming the evaluation setup is correct.

The associated evaluation framework is provided by **DOVE** [Chen et al., 2025] (NeurIPS 2025), the current VSR benchmark suite. DOVE combines the UDM10 and SPMCS test sets with a multi-metric evaluation framework that includes E*warp for temporal consistency. DOVE does not address long-video evaluation or multi-scale temporal metrics, and is therefore used in this proposal as a baseline framework rather than as a long-video evaluation target.

2.2 Temporal Consistency Metrics

Three families of temporal consistency metrics have been used in the video super-resolution and video generation literature. Each was originally designed for a different evaluation target; together they form the temporal-evaluation toolkit currently available to a super-resolution practitioner.

- **tOF and tLP** (TecoGAN, Chu et al. [Chu et al., 2020]) measure per-frame optical-flow consistency (tOF) and LPIPS-based perceptual warping error (tLP). Both are defined at a single adjacent-frame gap (k=1). The preliminary work in Section 5 demonstrates that tOF ordering inverts between k=1 and $k \geq 10$: the smoother method (UAV) wins at k=1 but loses to the detail-preserving method (MGLD-VSR) at long range.
- **E*warp** [Chen et al., 2025] extends warp-based error evaluation as part of the DOVE benchmark; it remains a k=1-dominant metric and shares the adjacent-frame limitation of tOF.
- **VBench** [Huang et al., 2023] and **VBench-2.0** [Huang et al., 2025] provide multi-aspect evaluation of generated video, including subject and background consistency (DINOv2 cosine similarity between adjacent frames), Human Identity (slow-fast ArcFace clip adapter), and Human Anatomy (anomaly ViT). The preliminary work documents applicability gaps for super-resolution evaluation: DINOv2-based consistency metrics exhibit smoother-output bias and the Human Anatomy dimension has content-dependent calibration failure on close-up footage.

2.3 No-Reference Perceptual Metrics

Three no-reference perceptual metrics are commonly applied to super-resolution outputs when high-resolution ground truth is unavailable. Each is single-frame or per-clip in scope and none specifically targets long-range cross-frame consistency, but together they form the no-reference perceptual evaluation toolkit relevant to deployment-time super-resolution assessment.

- **CLIP-IQA** [Wang et al., 2022] evaluates per-frame image quality using CLIP embeddings trained on diverse natural content. Unlike LPIPS or DINOv2, CLIP-IQA does not systematically penalise high-frequency detail in diffusion-style super-resolution outputs, making it a suitable component of a no-reference composite for super-resolution evaluation.
- **MUSIQ** [Ke et al., 2021] uses a multi-scale image quality transformer. The detail-preserving baseline scores higher than the text-conditioned smoother baseline on MUSIQ across all test videos in our preliminary work.
- **DOVER** [Wu et al., 2022] combines technical and aesthetic video quality dimensions through two disentangled convolutional branches whose outputs are combined into a single video-level score. The detail-preserving baseline leads the text-conditioned smoother by approximately +8.75 DOVER points on the per-method mean over the real-SR test set.

2.4 Identified Literature Gap

Three properties are needed for rigorous long-video SR evaluation but absent from any single published metric:

1. **No-reference** — HR ground truth is unavailable in deployment.
2. **Multi-time-scale** — adjacent-frame metrics miss the tOF crossover at k=5–10 frames documented in preliminary work.
3. **Bias-calibrated** — three independent learned-representation metrics (DINOv2, anomaly ViT, LPIPS) share a structural smoother-output bias; a principled aggregation must account for this.

The proposed Long-Range Video Consistency Composite (LR-VCC) metric, described in Section 6, is designed to satisfy all three.

3 Statement of the Problem

The evaluation of long-form video super-resolution is structurally insufficient in four overlapping ways, each of which is documented empirically in the subsections below.

First, **per-frame fidelity metrics** (PSNR, SSIM, LPIPS, DISTS, CLIP-IQA) are temporal-blind by construction: a video that drifts slowly across thousands of frames in colour, or that jumps at every tile boundary of a tile-based diffusion super-resolver, or whose face identity degrades from minute one to minute two, scores identically to a colour-stable, seam-free, identity-preserving video provided each individual frame is locally crisp.

Second, **short-range temporal metrics** (E*warp from DOVE [Chen et al., 2025], tOF and tLP at adjacent frame lag $k = 1$ from TecoGAN [Chu et al., 2020]) are scale-blind in the opposite direction. They measure adjacent-frame smoothness but cannot see consistency across the longer time horizons where the most damaging long-video failures actually occur. They also suffer from a systematic *smoother-output bias*: a spatially smoother output has mechanically lower frame-to-frame difference regardless of whether it is more consistent.

Third, **perceptual aggregates** (DOVER [Wu et al., 2022], CLIP-IQA trajectories, VBench-2.0 [Huang et al., 2025]) compute clip-level scores that are then averaged up to a per-video number. The averaging removes the very consistency signal we want to recover. VBench-2.0's long-video extension is in practice clip-level re-aggregation: the video is partitioned into short clips, each clip is evaluated by the same dimension-specific scorer, and the per-clip scores are averaged with no measurement of cross-clip drift.

Fourth, and most subtly, **even metrics that should catch a failure mode have content-dependent calibration shifts** that are not surfaced to the user. The same metric, computed under identical hyperparameters on two videos of comparable overall quality, can flip its method ranking on one of them because some property of the content (close-up faces, dense motion, dim lighting) pushes a learned classifier into a different operating regime. The metric returns a score; it does not return "I am out of distribution on this video."

These four breakdowns reinforce each other. A researcher who wishes to know whether method A is more temporally consistent than method B on long-form content has, today, no single trustworthy answer. The remainder of this section documents each breakdown empirically.

3.1 A Content-Dependent Regime Shift in VBench-2.0 Metrics

Two slow-fast-adapted VBench-2.0 metrics — Human_Anatomy (anomaly classifier on per-frame body / face / hand detector ensemble) and Human_Identity (RetinaFace + ArcFace face embedding stability across the video) — were evaluated on a real-SR test set comprising five long-form synthetic super-resolution videos (320×180 low resolution, 1280×720 super-resolved, 22,412 frames in total). Both metrics were adapted to long videos through a 2-second sliding-clip protocol with slow (within-clip) and fast (cross-clip first-frames) components fused.

On four of the five videos, both metrics rank a detail-preserving diffusion baseline higher than a text-conditioned smoother baseline, in agreement with informal viewing. On the fifth video — a close-up scene where the smoother method's output was visually substantially worse — **both metrics flip rankings in disagreement with the visual evidence**. Two independent metrics failing in the same direction on the same video, on the one video where the visual judgement is most confident, cannot be dismissed as evaluation noise.

Per-frame Anatomy fire-rate traces decomposed by detector channel and anomaly threshold identify the fraction of frame area occupied by the largest detected hand bounding box as the regime-shift trigger. Across the five-video cohort the median hand bounding-box fraction is approximately 2–3% of frame area; on the regime-shift video it is approximately 18%. The anomaly classifier was trained on a video distribution where close-up hands are rare; on close-up content it fires more aggressively, and it fires *more aggressively on sharper output* because sharper edges produce stronger detector activations. The classifier therefore inverts on close-up content: the visually better method is flagged as more anomalous by the trained behaviour the classifier is supposed to provide. Human_Identity exhibits a parallel inversion on the same video, likely through face-detection-confidence cliffs at close range.

The generalised lesson is that VBench-style anomaly-detector metrics have content-dependent calibration shifts that the metric does not surface. A video whose content happens to push the classifier into the high-fire regime will see every method evaluated on it scored under different effective calibration than other videos. There is no per-video reliability signal in the standard pipeline that flags "this video is out of the metric's calibrated range." The proposed metric (Section 6) addresses this by introducing per-sub-metric *reliability gates* that downweight any sub-metric whose calibration assumptions are violated on the video at hand.

3.2 The tOF Temporal-Scale Crossover

The tOF metric [Chu et al., 2020] measures temporal consistency through optical-flow warping error and is conventionally evaluated at adjacent-frame lag $k = 1$. The framework was extended to compute tOF and tLP at multiple lags $k \in \{1, 5, 10, 30, 60, 120\}$ using RAFT-based optical flow with forward-backward consistency masking applied uniformly across all k.

The winner ranking flips with k. At $k = 1$ the text-conditioned smoother baseline wins on four of five videos, because adjacent-frame transitions of a spatially smoother

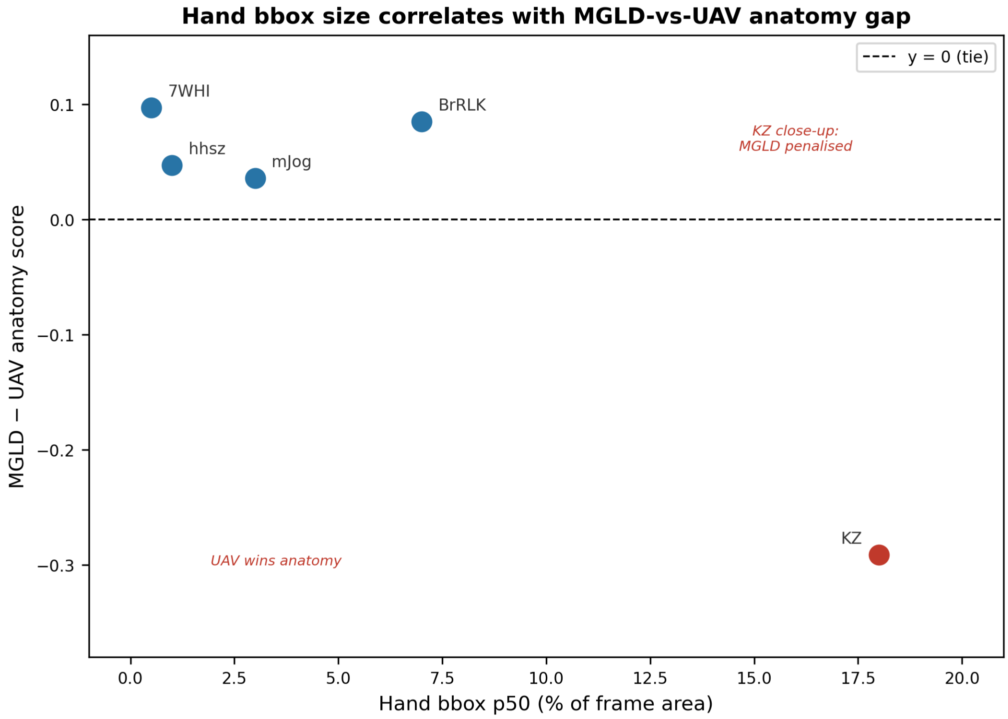

Figure 1: Anatomy metric difference between the detail-preserving and the text-conditioned smoother baselines, plotted against per-video median hand-bounding-box fraction. Four videos cluster in the upper-left at low close-up fraction, with the detail-preserving baseline correctly winning. The regime-shift video sits alone in the lower-right at approximately 18% hand-bbox fraction — well above the cohort median of 2–3% — with the metric inverted by approximately -0.29. The pattern is consistent with a content-dependent calibration shift triggered by close-up scale.

output have mechanically lower warping error. At $k \geq 30$ the detail-preserving diffusion baseline wins on four of five videos, because the smoother method's per-frame smoothness comes at the cost of cumulative drift across longer time horizons — drift that long-range warping picks up directly. The crossover point lies at $k \approx 5$–10; tLP exhibits the same crossover at the same range.

The interpretation is that short-range temporal metrics suffer from the smoother-output bias in its purest form: a smoother output has less to differ from the previous frame, so frame-to-frame difference is mechanically lower. This same bias affects DOVER's aesthetic branch, DINOv2-cosine cross-frame similarity, and the Anatomy fire-rate. The literature convention of reporting tOF at $k = 1$ as the temporal-consistency proxy is, in effect, a spatial-smoothness measurement.

The practical consequence is that a study that claims a temporal-consistency improvement using tOF at $k = 1$ may be measuring spatial smoothness, not consistency. A method that wins on the standard benchmark by lowering tOF at $k = 1$ may lose to its competitor at $k = 60$. Neither metric is correct in isolation; the picture only resolves when k is varied. The proposed metric (Section 6) aggregates tOF across $k \in \{1, 5, 10, 30, 60, 120\}$ with a weighting that gives meaningful representation to every temporal scale.

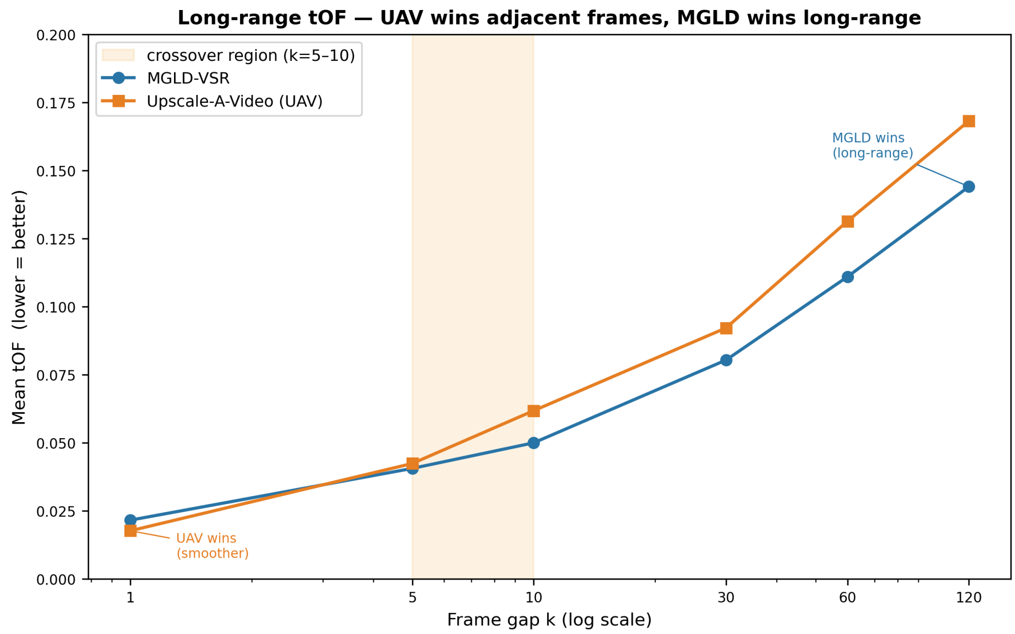

Figure 2: Mean tOF as a function of frame gap k for the two super-resolution baselines. At $k = 1$ the text-conditioned smoother baseline (orange) scores lower, consistent with the smoother-output bias of adjacent-frame temporal metrics. At $k \geq 30$ the detail-preserving baseline (blue) scores lower, registering the smoother baseline's cumulative long-range drift. The crossover region (shaded) lies at $k = 5$–10.

3.3 A Synthetic Artefact Battery Exposes Categorical Blind Spots

To establish the empirical reach of the four limitations above, a parameterised synthetic artefact pipeline was constructed that operates on existing super-resolved outputs and injects controlled degradations at known severities. Four artefact families were implemented:

- **Colour drift** — a linear ramp: at frame i of T total frames, the red channel is multiplied by $1 + \alpha \cdot i/T$ and the green and blue channels are reduced by the same factor. Severity $\alpha \in \{0.02, 0.05, 0.10, 0.20, 0.40\}$. Tests detection of slow monotonic drift across the video.
- **Chunk-boundary jumps** — at every 60th frame ($\approx$ two seconds at 30 fps), a deterministic per-chunk uniform additive offset of magnitude $\pm\alpha \cdot 255$ is applied to all channels. Severity controls step amplitude. Mimics the seams produced by tile-based diffusion super-resolution.
- **Periodic flicker** — sinusoidal brightness modulation: per-frame intensity is multiplied by $1 + \alpha \sin(2\pi i/P)$ with period $P = 15$ frames. Severity controls peak amplitude. Mimics the periodic flicker produced by some diffusion samplers.
- **Identity degradation** — per-frame frontal-face detection followed by a Gaussian blur with $\sigma = 10\alpha$ applied within each detected face bounding box. Severity controls blur sigma. Mimics the identity-collapse failure mode of diffusion super-resolution on faces.

Each generator is verified by unit tests checking that severity zero leaves the video unchanged within floating-point tolerance, that per-frame statistics match the analytic target curve, and that the construction is seed-reproducible.

Eight baseline metrics from the super-resolution literature were evaluated on the artefact videos: CLIP-IQA, tOF at $k = 1$, tOF at $k = 60$, tOF at $k = 120$, tLP at $k = 120$, E*warp, DOVER, and Identity (fused slow-fast). For each (metric, artefact family) cell, the test is whether the metric responds monotonically across the five severities on both base videos.

Of the resulting 32 cells (8 metrics $\times$ 4 artefact families), only **five cells are clean PASS**. The verdict matrix is presented in Table 1.

Table 1: Severity-response verdicts of eight widely-used baseline metrics across four synthetic artefact families. PASS denotes monotonic severity-response on both base videos; PARTIAL denotes monotonic on one base or response only as a global side-effect; FAIL denotes flat or non-monotonic response.

Metric	Colour drift	Chunk-boundary	Flicker	Identity deg.
CLIP-IQA	FAIL	FAIL	FAIL	PARTIAL
tOF ($k = 1$)	FAIL	FAIL	**PASS**	FAIL
tOF ($k = 60$)	FAIL	**PASS**	FAIL	FAIL
tOF ($k = 120$)	FAIL	**PASS**	FAIL	FAIL
tLP ($k = 120$)	FAIL	**PASS**	FAIL	FAIL
E*warp	FAIL	**PASS**	FAIL	FAIL
DOVER	FAIL	FAIL	FAIL	FAIL
Identity (fused)	FAIL	FAIL	FAIL	PARTIAL

Four headline findings follow from this matrix.

Finding 1 — Colour drift is a categorical blind spot of the entire baseline metric suite. No baseline metric responds monotonically across the severity range on either base video. The mechanism is structural: a slow linear colour ramp produces a per-pair colour shift between frames k apart of $\alpha k/T$. For the conditions tested this per-pair shift sits below the histogram bin width of conventional colour-histogram metrics, below the noise floor of optical-flow warping error (the flow estimator absorbs uniform colour shifts), and below the perceptual threshold of CLIP-IQA and DOVER (the per-frame quality of a slightly red-shifted frame is indistinguishable from the original). The signal exists in the trajectory of per-channel means across the entire video, not in any per-pair comparison.

Finding 2 — Chunk-boundary detection requires temporal lag k at least as large as the chunk size. All four long-range temporal metrics catch chunk-boundary cleanly. Per-frame metrics, the adjacent-frame temporal metric, and the clip-level perceptual metrics are entirely blind. This proves that a single-scale temporal metric, however well-tuned, will miss artefacts whose characteristic frequency does not match its measurement scale.

Finding 3 — Flicker is caught only at the smallest temporal lag. Long-range temporal metrics ($k \geq 30$) are blind to periodic flicker because the flicker period of 15 frames divides their measurement scale cleanly ($60 = 4P$, $120 = 8P$), so the per-pair difference cancels by construction. tOF at $k = 1$ catches it cleanly — the response is monotonic with severity ratio of approximately 4.5$\times$ from the lowest to the highest severity. This is the converse of Finding 2: aggregation at long temporal scales alone is insufficient.

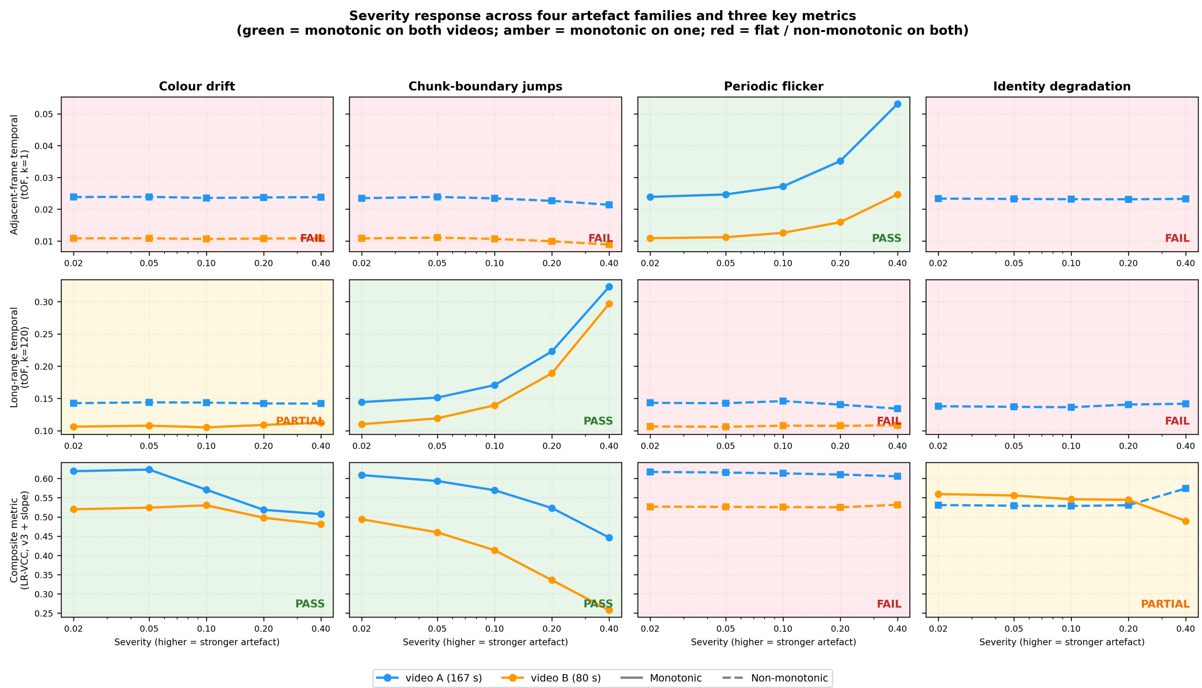

Figure 3: Severity response across the four artefact families and three representative metrics from the baseline suite: adjacent-frame temporal (tOF at $k = 1$), long-range temporal (tOF at $k = 120$), and the proposed composite (LR-VCC v3 + slope). Each panel plots metric value against artefact severity on the two base videos. Green panels denote monotonic severity-response on both videos; amber panels denote monotonic on one video; red panels denote flat or non-monotonic response on both. The pattern of colour shows that no single baseline metric catches every artefact family, while the composite catches the artefact families whose mechanisms are addressed by its constituent sub-metrics and degrades gracefully on the others.

Finding 4 — Identity degradation triggers a content-dependent inversion of the slow-fast Identity metric. On the multi-face base video the Identity sub-metric responds correctly (score drop of approximately 0.23 across the severity range). On the single-face base video the same sub-metric *rises* with severity by approximately 0.11, because heavy face-region blur ($\sigma = 4.0$ at maximum severity) erases identity-distinctive features in all frames of the single face, so cross-clip first-frame embeddings become more similar in a "generic blurred face" sense. The face detector survives the blur (face detection rate of approximately 0.96 at every severity), so face-rate-based reliability gating does not engage. The pathology lies in the slow-fast pooling itself.

3.4 Why the Gaps Are Structural

The blind spots above are not artefacts of a particular threshold choice or training-data deficit. Each has a structural cause: per-frame metrics have no temporal awareness by construction; adjacent-frame temporal metrics exhibit the smoother-output bias by construction; clip-level perceptual aggregates remove the cross-clip signal by construction; per-pair temporal at a fixed k is scale-blind to artefacts whose characteristic frequency does not match k; and VBench-style anomaly-detector metrics have content-dependent calibration shifts intrinsic to learned classifiers operating on a wider content distribution than they were trained on.

Progress requires (a) multi-scale temporal sampling that gives weight to both high-frequency and long-range failure modes, (b) an explicit colour-trajectory sub-metric that catches drift slower than any per-pair measurement scale, (c) reliability gating per sub-metric so that content-dependent metric flips suppress themselves on the videos where they apply, and (d) a composition layer that downweights unreliable sub-metrics on-the-fly rather than reporting a single fixed metric. These four requirements specify the design space for any proposed long-video consistency metric; the metric introduced in Section 6 satisfies each.

4 Research Aim and Hypothesis

4.1 Research Aim

The aim of this research is to develop and validate a no-reference long-range video consistency metric for long-form video super-resolution that satisfies three properties. First, the metric catches the long-range failure modes documented in Section 3 to which existing metrics are demonstrably blind: slow colour drift, chunk-boundary jumps from tile-based diffusion, periodic flicker from sampler artefacts, and identity collapse on recurring faces. Second, the metric preserves correct per-video method rankings on real super-resolved outputs, including in the regime-shift case (Section 3.1) where two VBench-2.0 metrics flip rankings in disagreement with human visual judgement. Third, the metric is reproducible, modular, and extensible, so that additional sub-metrics can be added as failure modes are characterised without disturbing the existing composition.

The metric introduced for this purpose is the **Long-Range Video Consistency Composite (LR-VCC)**. Its specification and design rationale are given in Section 6; its validation strategy is given in Section 6 together with the method.

4.2 Research Hypotheses

The research is organised around three falsifiable hypotheses. Each is testable against the synthetic artefact set introduced in Section 3.3 and against the real super-resolution test set.

Hypothesis H1 (Composition). *Reliability-weighted softmax-log-mean composition of independent sub-metrics catches failure modes that no single constituent sub-metric catches.*

For each artefact family, the per-video severity-response of each sub-metric individually and of the composite is computed. The hypothesis is supported if the composite catches at least one artefact family whose detection requires sub-metric combination — specifically, a family where the colour and temporal sub-metrics provide independent and complementary evidence and where neither alone produces a monotonic severity-response.

Hypothesis H2 (Multi-scale temporal). *Aggregating temporal warping error over multiple frame gaps catches both high-frequency and long-range temporal failure modes that fixed-scale temporal metrics miss.*

The same artefact set, restricted to flicker (high-frequency) and chunk-boundary (mid-range), is used to compare fixed-scale tOF at $k = 1$ against the proposed multi-scale aggregation under different weighting choices over $k \in \{1, 5, 10, 30, 60, 120\}$. The hypothesis is supported if the multi-scale aggregation catches both artefact families at the temporal sub-metric level (the corresponding composite-level behaviour is studied separately, since composite-level behaviour depends additionally on the reliability gates of the other sub-metrics).

Hypothesis H3 (Reliability gating against content-dependent flips). *Per-sub-metric reliability gates derived from interpretable signals (face-detection rate, close-up content indicator, regression goodness-of-fit, histogram entropy, and optical-flow mask coverage) reduce content-dependent ranking flips of the type observed in the regime-shift case study.*

The composite is evaluated on the real super-resolution test set, including the regime-shift video. The hypothesis is supported if the composite preserves a consistent ranking across all test videos with a per-video difference between the two methods that remains positive on every video and is substantially above the noise floor of the constituent sub-metrics in expectation.

4.3 Success Criteria

The research objectives are met if the following measurable outcomes are achieved over the proposal-to-thesis horizon.

- The composite catches a majority of (artefact family, base video) conditions monotonically. The remaining conditions are characterised with identified failure mechanisms that become future-work targets, not unexplained anomalies.
- The composite preserves correct per-video method rankings on every video of the real super-resolution test set, including the regime-shift case.
- Every sub-metric is reliability-gated by an interpretable, decomposable signal, not a black-box weighting.
- Every iteration of the composite closes a specific failure mode characterised on the validation set, not an unmotivated parameter tweak.
- The metric is reproducible from a single canonical configuration, with cached per-video sub-metric values so that hyperparameter studies do not require rescanning videos.

5 Research Objectives, Content, and Preliminary Work

To achieve the aim and test the hypotheses stated in Section 4, this research is organised around three objectives. The work completed to date provides a partial validation of each.

5.1 Objective 1: Characterise the Failure Modes of Existing Long-Video SR Metrics

The first objective is to establish, empirically, the precise failure modes of widely-used super-resolution evaluation metrics on long-form video. The findings of this objective motivate the design choices of the proposed metric (Section 6); they are also a contribution in their own right because they document gaps in the existing metric literature that have not previously been systematically characterised.

Real-SR test set. A test set of five long-form synthetic super-resolution videos has been prepared, each at 1280×720 output resolution, with total frame counts ranging from 2,400 to 5,000 per video and durations from 80 to 208 seconds at native frame rates. The total frame count across the set is 22,412. Each test video was generated by applying the DOVE [Chen et al., 2025] degradation pipeline (bicubic downscale by a factor of four, sensor noise injection, and codec compression) to a real source clip, and then upscaled by two representative super-resolution baselines: a detail-preserving diffusion-based method (MGLD-VSR [Yang et al., 2023]) and a text-conditioned temporal-diffusion smoother (Upscale-A-Video [Zhou et al., 2023]). Before drawing conclusions from any metric values, the inference environment was validated against published DOVE benchmark numbers: the detail-preserving baseline reproduces the published UDM10 results exactly to four decimal places (PSNR, SSIM, LPIPS, DISTS); the smoother baseline reproduces them within a consistent positive offset attributable to a minor degradation pipeline configuration difference rather than a setup defect. The thesis is therefore evaluated on a numerically validated baseline pair; extension to a wider set of recent super-resolution methods is identified as planned post-proposal work.

Established findings. Three principal findings from the failure-mode characterisation are described in Section 3 and summarised here:

1. A regime-shift case in which two VBench-2.0 metrics (Human_Anatomy and Human_Identity) flip rankings in disagreement with visual judgement on a close-up scene, traceable to a content-type predictor (hand bounding-box fraction relative to the cohort median) that pushes the trained anomaly classifier into a high-fire regime (Section 3.1).
2. A temporal-scale crossover in which the winner identified by tOF flips between $k = 1$ and $k \geq 30$, demonstrating that adjacent-frame temporal metrics measure spatial smoothness rather than consistency (Section 3.2).
3. A controlled severity-response analysis on a four-artefact synthetic test set showing that of 32 (metric, artefact family) cells across eight widely-used baseline metrics, only five cells PASS, two PARTIAL, and the remaining 25 FAIL — with slow colour drift in particular invisible to every baseline metric (Section 3.3).

A subsidiary characterisation, also completed but not detailed here, is the discovery and correction of a frame-rate metadata inconsistency in both super-resolution pipelines: each baseline tagged its output at a hard-coded frame rate independent of the source frame rate, which introduced a small but systematic bias in any per-second temporal aggregation. The correction is incorporated into all subsequent measurements.

5.2 Objective 2: Design a No-Reference Composite Metric with Per-Sub-Metric Reliability Gates

The second objective is to design a no-reference composite metric whose construction is *driven by* the failure characterisations of Objective 1. The metric must include sub-metrics that catch the specific failure modes characterised, must aggregate them with a composition layer that downweights unreliable signals per video, and must be modular so that new sub-metrics can be added as new failure modes are characterised without disturbing the existing composition.

Five sub-metrics have been designed and validated to date: Appearance (CLIP-IQA), Temporal (multi-scale optical-flow warping error), Identity (slow-fast face embedding adapter), Colour Stability (Lab-channel histogram L1 distance over multi-scale pairs), and Colour Trajectory Slope (linear regression on per-frame Lab-channel means with goodness-of-fit-gated reliability). Their specification is in Section 6.

5.3 Objective 3: Validate the Composite Against a Controlled Synthetic Test Set and the Real-SR Test Set

The third objective is to validate the composite metric at two layers. The first layer is perceptual agreement on the real super-resolution test set: the composite must rank the detail-preserving baseline higher than the text-conditioned smoother on every test video including the regime-shift case. The second layer is severity-response on the synthetic artefact set: the composite must respond monotonically to severity for each artefact family on at least one base video, with any cases of non-monotonicity attributable to a *named* failure mechanism rather than to unexplained noise.

Both validation layers have been carried out for the current production version of the composite. Layer-one validation passes on every test video including the regime-shift case, with a mean difference of approximately +0.06 in favour of the detail-preserving baseline. Layer-two validation passes on the majority of the (artefact family, base video) cells; the remaining cells have identified, mechanism-named failure modes which are recorded as future-work targets in Section 7, together with a planned extension of the validation to a broader set of recent super-resolution methods.

6 Methodology and Technical Route

This section specifies the proposed Long-Range Video Consistency Composite (LR-VCC) metric, its design rationale, and the iterative protocol under which the current production version was derived. The metric is constructed directly from the four failure modes characterised in Section 3; each sub-metric closes a specific gap, and the composition layer downweights any sub-metric whose calibration assumptions are violated on the video under evaluation.

6.1 Architecture

Five sub-metrics, indexed by $s \in \{A, T, I, D, E\}$, are computed in parallel from a super-resolved video V:

- **Sub-metric A — Appearance stability.** Per-frame CLIP-IQA [Wang et al., 2022] is computed over the video. The score is the mean per-frame quality minus a small multiple of its standard deviation across frames, rewarding outputs that are not only high-quality on average but consistently so across minutes of footage. The mean term rewards perceptual quality; the standard deviation term penalises temporal instability of perceptual quality.

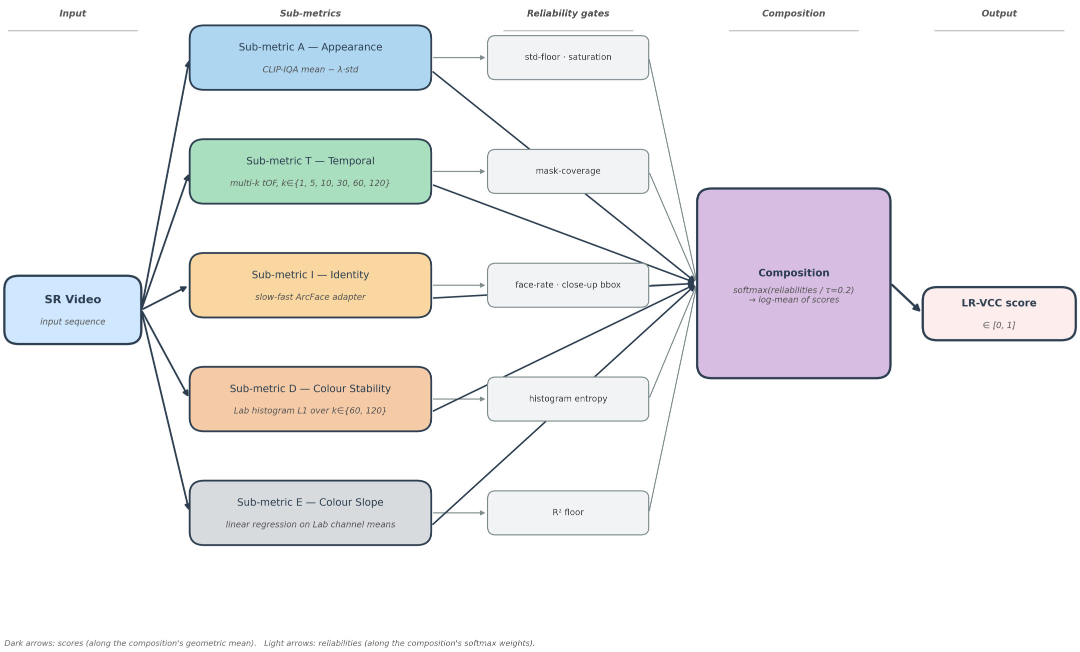

Figure 4: LR-VCC architecture: a super-resolved video as input; five sub-metrics computed in parallel, each emitting a (score, reliability) pair in $[0, 1] \times [0, 1]$; a reliability-weighted softmax-log-mean composition produces the final LR-VCC score in $[0, 1]$.

- **Sub-metric T — Temporal stability.** Optical-flow warping error (tOF [Chu et al., 2020]) is computed at multiple temporal lags $k \in \{1, 5, 10, 30, 60, 120\}$ using RAFT-based optical flow with forward-backward consistency masking. The aggregate is a weighted mean over k, with the choice of weighting (logarithmic, uniform, or square-root) configurable; uniform weighting is adopted as the production setting because the empirical temporal-scale crossover characterised in Section 3.2 shows that no single k is sufficient. The score is one minus the (normalised) weighted mean warping error.

- **Sub-metric I — Identity preservation.** The slow-fast Human_Identity adapter (RetinaFace + ArcFace face embedding stability across 2-second clips with both within-clip and cross-clip first-frame components) provides the score, with reliability gated by the face-detection rate (a video with few detected faces does not carry an identity signal) and by a close-up content indicator (the same trigger identified in Section 3.1) that downweights the sub-metric on videos in the regime where the underlying anomaly classifiers exhibit content-dependent calibration shifts.

- **Sub-metric D — Colour stability.** Per-frame Lab-channel histograms are compared at L1 distance over multi-scale frame pairs $k \in \{60, 120\}$. The score is an exponential decay in the mean pair-distance, with the decay constant calibrated empirically so that clean super-resolved outputs score in the middle of the dynamic range. Reliability is gated by histogram entropy — a video whose histogram is too flat carries no discriminative colour information.

- **Sub-metric E — Colour trajectory slope.** Per-frame Lab-channel means are fit by linear regression as a function of frame index. The score is an exponential decay in the maximum absolute slope across channels. Reliability is the maximum coefficient of determination (R^2) across the three channels, gated by an R^2 floor. A drifting video has high R^2 (the slope is real, so the metric speaks confidently); a flickering video has R^2 near zero (sinusoidal residuals dominate, so the metric correctly abstains); a clean video has near-zero slope (the score is near one regardless of reliability).

Each sub-metric outputs a score $s_i \in [0, 1]$ and a reliability $r_i \in [0, 1]$. The composition is a reliability-weighted log-mean:

$$w_i = \frac{\exp(r_i/\tau)}{\sum_j \exp(r_j/\tau)}, \tag{1}$$

$$\text{LR-VCC}(V) = \exp\left(\sum_i w_i \log(s_i + \varepsilon)\right), \tag{2}$$

with temperature $\tau = 0.2$ (a sharp softmax: the most reliable sub-metric dominates) and $\varepsilon = 10^{-6}$ for numerical stability. A video for which all five reliabilities fall below a low-confidence threshold is flagged in the output and excluded from per-method aggregates by default.

The log-mean composition preserves the no-compensation property across sub-metrics: a sub-metric scoring near zero pulls the composite down unless the other sub-metrics carry overwhelming weight, which in turn requires the failing sub-metric's reliability to be near zero. A weighted arithmetic mean would not have this property: a single sub-metric failure could be diluted away by averaging.

6.2 Reliability Gating as the Core Design Pattern

The central design hypothesis is that every existing learned-representation video metric is reliable on some content and unreliable on other content; rather than discard failed metrics or attempt to train new bias-free metrics, the composition uses existing metrics honestly and controls their influence per video through reliability scores derived from the regime indicators that predict their failure mechanisms.

Each per-sub-metric reliability is computed from regime indicators using smooth sigmoid penalties (sharpness fixed at 10, producing smooth transitions rather than sharp threshold cliffs). The thresholds were selected from the regime characterisations in Section 3; they are not tuned to the validation set.

A sharp softmax over reliabilities ensures that when one sub-metric is substantially more trustworthy than the others on a given video, that sub-metric clearly dominates the composition. This is the mechanism that makes the composite resistant to the content-dependent ranking flips of Section 3.1: on a video in the close-up regime, where the Identity-pathway anomaly classifier is known to misbehave, the Identity sub-metric's reliability is downweighted by the close-up gate, allowing the other four sub-metrics to determine the composite outcome.

6.3 Validation Strategy

The composite is validated at three layers, as introduced in Section 5.3.

Layer 1 — perceptual agreement on the real-SR test set. The composite is evaluated on the real super-resolution test set comprising both super-resolution baselines. The pass criterion is that the detail-preserving baseline wins on the per-method mean and on every per-video comparison. The current production version meets this criterion: the per-video comparison is consistent across all test videos including the regime-shift case, with a mean difference between the two methods of approximately +0.06 in favour of the detail-preserving baseline.

Layer 2 — flip-resistance on the regime-shift case. The composite must not flip on the close-up video where Human_Anatomy and Human_Identity both flip. The criterion is met by the design: sub-metric I's reliability gate engages on close-up content, sub-metric T uses warping error directly (not LPIPS-based tLP, which exhibits the smoother-output bias of its underlying perceptual representation), and sub-metric A is built on CLIP-IQA, which does not depend on a pristine-HR-trained representation.

Layer 3 — severity-response on the synthetic artefact set. For each of the four artefact families, the composite is expected to respond monotonically to severity on at least one base video. The current production version catches the majority of (artefact family, base video) cells monotonically. The remaining cells have characterised failure modes:

- On the single-face base video, identity degradation triggers the identity-collapse pathology of Section 3.3 Finding 4. The face-detection rate remains high, so the existing face-rate reliability gate does not engage; the pathology lies in the slow-fast pooling itself. A reliability gate that operates on per-face embedding variance is identified as the natural fix (Section 7).

- Flicker is correctly detected by sub-metric T at small k, but the composite is dominated by the colour and identity sub-metrics whose reliabilities are near one and whose scores do not respond to high-frequency luminance modulation; the composite is therefore approximately flat across flicker severities. A fast-varying brightness sub-metric is the natural addition.

- One of the two base videos for colour drift exhibits a pre-existing slow colour trajectory in the underlying super-resolved output. The slope-style sub-metric correctly assigns high reliability to that pre-existing trajectory, leaving little additional dynamic range to register the injected drift. The case is informative rather than spurious.

The full verdict matrix and per-sub-metric severity-response curves are presented in Figures 5–7.

6.4 Reproducibility and Decomposability

The metric is fully decomposable. Each sub-metric is computed independently, and the result is cached per video. Hyperparameter studies and re-tuning of the per-sub-metric decay constants and weighting choices do not require re-scanning videos; the cached per-video values are re-aggregated through the composition layer in seconds. Additional sub-metrics can be added to the composition without disturbing the cached values of the existing sub-metrics.

The output of the composite on a given video is the LR-VCC score, the per-sub-metric $(\text{score}, \text{reliability})$ pairs, the softmax-derived sub-metric weights, the low-confidence flag, and a set of per-sub-metric diagnostic indicators (closeup indicator, face-detection rate, mean mask coverage at long temporal lags, regression R^2, histogram entropy). The per-video per-sub-metric matrix is always reported alongside the headline composite, so a reviewer can see where the composite is driven from on each video. There is no black-box aggregation.

7 Expected Outcomes and Innovations

This research is expected to deliver three artefacts and four scientific contributions, each described in turn.

7.1 Primary Deliverables

The first deliverable is the proposed metric itself: a no-reference long-range video consistency composite for super-resolution evaluation, with five reliability-gated sub-metrics covering appearance, multi-scale temporal consistency, identity preservation, colour-distribution stability, and colour-trajectory drift. The composite is decomposable: the per-sub-metric scores and reliabilities are reported alongside the composite so that each video's evaluation is traceable to the sub-metric driving it. The composite is fully specified by a small set of canonical hyperparameters; alternative hyperparameter choices can be explored through cached per-video sub-metric values without rescanning videos.

The second deliverable is a parameterised synthetic test set for long-video super-resolution consistency assessment, covering four artefact families — slow colour drift, chunk-boundary jumps, periodic flicker, and identity degradation — across multiple severity levels per base video. Each generator is verified to produce zero output at zero severity (identity within floating-point tolerance), monotonic per-frame statistics under severity sweep, and seed-reproducible construction. The synthetic test set is the first publicly characterised consistency benchmark targeting long-video super-resolution specifically; existing video super-resolution benchmarks (REDS, Vimeo-90K, Vid4, UDM10, SPMCS) target per-frame fidelity on short clips. The test set is extensible: additional base videos, finer severity grids, and new artefact families can be added without disturbing the existing infrastructure.

The third deliverable is the empirical-characterisation contribution of Section 3. The four findings — the regime-shift case, the temporal-scale crossover, the colour-drift blind spot of the baseline metric suite, and the identity-collapse pathology of slow-fast pooling under heavy degradation — are documented, each with an identified root mechanism. To our knowledge these characterisations have not previously been compiled into a single systematic study of long-video super-resolution evaluation.

7.2 Scientific Innovations

The research introduces four innovations.

Reliability-weighted softmax-log-mean composition as a design pattern for no-reference long-video metrics. The composition assigns weights through a sharp softmax over per-sub-metric reliabilities, then computes a geometric mean of scores under those weights. Sub-metrics whose reliability gates engage on a given video are down-weighted automatically, without per-sub-metric threshold-based hard exclusion. The composition is sub-metric-agnostic: additional sub-metrics can be added as new failure modes are characterised without disturbing the existing composition. To our knowledge this is the first long-video super-resolution consistency metric of this form.

Goodness-of-fit-gated reliability for trajectory-style sub-metrics. Sub-metric E addresses the colour-drift blind spot by fitting a linear regression to per-frame Lab-channel means over the whole video, scoring an exponential decay in the maximum absolute slope, and gating reliability by the maximum coefficient of determination across the three channels. A drifting video produces a high R^2 (the slope is real, so the metric speaks confidently); a flickering video produces R^2 near zero (sinusoidal residuals dominate, so the metric correctly abstains). The pattern — gate reliability by goodness-of-fit, not by score magnitude — is reusable for any slow-signal-versus-noisy-baseline detection problem in video evaluation.

Multi-scale temporal aggregation with configurable weighting. The proposed sub-metric T aggregates tOF over $k \in \{1, 5, 10, 30, 60, 120\}$; the per-video per-$k$ warping errors are cached so that alternative aggregation weightings can be applied at composition time without recomputing the underlying optical flow. This enables principled hyperparameter sweeps over the temporal-scale axis that produced the crossover finding of Section 3.2.

8 Main References

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